

Tran Manh Duy

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Summary

Systems-oriented Rust developer with a strong foundation in Computer Science and low-level programming. Experienced in building high-performance, low-latency services with a deep understanding of systems programming concepts including memory management, concurrency, and inter-process communication (gRPC, message queues). Proficient in designing modular, trait-based architectures and real-time event processing pipelines using Kafka and async Rust (Tokio). Eager to apply systems-level expertise to security infrastructure and real-time telemetry platforms.

Education

University of Engineering and Technology, VNU

BS in Information Technology, CPA: 3.0/4.0

Available for Full-time employment

Hanoi, Vietnam

Sept 2022 – Sept 2026

Skills

Programming Languages: Rust (Advanced), C/C++ (Intermediate), Golang, Typescript

Systems & Concurrency: Tokio (async runtime), memory management (ownership, lifetimes, unsafe), concurrency primitives, trait-based design patterns, IPC (gRPC, message queues)

Frameworks & Protocols: Axum, Actix, Tonic (gRPC), Prost (Protocol Buffers)

Infrastructure: Kafka, RabbitMQ, Redis, Docker, Kubernetes, Linux, Git/GitHub, CI/CD (GitHub Actions)

Database: PostgreSQL, MySQL, MongoDB, SeaOrm, DieselOrm

Cryptography & Security: Custom ECC implementation, digital signatures (Schnorr, ECDSA), Montgomery field arithmetic, ZKP (zk-SNARKs, zk-STARKs)

Testing: Unit/Integration testing, Testcontainers, Mockall, load testing (k6), benchmarking (Criterion)

Language: English, Japanese

Experience

Sotatek ., JSC

Hanoi, Vietnam | Jan 2024 – Present

Rust Developer

- Designed and implemented high-performance **RESTful APIs** (using Axum/Actix) and **gRPC services** (using Tonic, Prost) to power CEX/DEX platforms, serving both web and mobile interfaces with low latency.
- Engineered scalable asynchronous messaging systems using **Kafka** and **RabbitMQ** to handle high-frequency trading data and real-time blockchain event processing.
- Optimized database performance and data reliability by integrating **Redis** for caching and **SeaOrm** for type-safe interactions with PostgreSQL.

Blockchain Developer

- Developed and deployed a Starknet Layer 2 solution integrated with Aptos Layer 1, including data availability on Move, Madara chain configuration, and a Rust-based prover for on-chain verification.

Research Assistant

- Researched, documented, and implemented algorithms related to Zero-Knowledge Proofs (ZKPs), including zk-SNARKs and zk-STARKs.

Personal Projects

High-Performance Trading Engine

- **Stack:** Rust, Cargo Workspace (hexagonal architecture), Criterion benchmarks.
- Designed and implemented a **low-latency limit order book** achieving **sub-microsecond** per-operation speed on isolated CPU cores, using cache-optimized data structures.
- Architected the system with **compile-time enforced layer boundaries** (LMAX Disruptor pattern), separating domain logic from infrastructure for deterministic **single-threaded event processing**.
- Implemented **concurrent-safe** order matching with lock-free structures and precise memory layout control for L1 cache efficiency.

High-Performance E-commerce Microservice

- **Stack:** Rust (Axum, Tokio), PostgreSQL (SeaOrm), Redis, Docker, GitHub Actions, k6.
- Architected a scalable backend using **Clean Architecture** and **Cargo Workspace** pattern, decoupling business logic (**core**) from infrastructure (**infra**) with **trait-based port/adaptor** interfaces.
- Solved critical **race conditions** in inventory management ensuring data integrity under high concurrency (verified via Testcontainers integration tests).
- Achieved **8,000 RPS** with 4.51ms avg latency (P95: 7.61ms) through **Redis caching** strategies and **load testing** (k6).
- Established a complete **CI/CD pipeline** using GitHub Actions for automated testing and Docker containerization.

Curve1024 – From-Scratch ECC Cryptography Library & CLI Tool

- **Stack:** Rust, custom 1024-bit BigNum (U1024), Montgomery field arithmetic, Schnorr/ECDSA.
- Implemented a complete **1024-bit Elliptic Curve Cryptography** library from scratch with zero external crypto dependencies, providing 256-bit symmetric security.
- Built custom **memory-safe large integer arithmetic** (U1024) and **Montgomery multiplication** for optimized modular field operations, demonstrating deep understanding of **low-level memory layout and bitwise operations**.
- Developed a **GPG-like CLI tool** (curve1024-sig) supporting key generation, Schnorr and ECDSA digital signatures, and file signing/verification.

2D Game Engine (C++ / SDL2)

- **Stack:** C++, SDL2, Makefile, state machine architecture.
- Built a 2D game engine in **C++** with modular separation (**src/ + include/** convention), implementing render loops, physics, texture management, and input handling.
- Designed a **state machine** architecture for game flow with clean module boundaries (Core, Characters, Physics, Map, Widget).
- Managed **manual memory allocation** and resource lifecycles (textures, sounds, SDL surfaces) in a non-GC environment.